

ELASTIC WAVES THROUGH A PACKING OF SPHERES*

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ABSTRACT

Based on a theory of porous solids previously developed by the author, the elasticity of a hexagonal close packing of equal spheres is treated. The packing is anisotropic and because of the weight of the spheres, also inhomogeneous. The velocities of propagation of elastic waves have been calculated for evacuated interspaces and for interspaces filled with a liquid or gas. In the case of evacuated or air-filled interspaces, the wave rays and travel times have been computed. The packing which has been treated may be of use as a model for a dry or wet loose material such as gravel or sand. Though the model is very simplified, the results obtained show some typical effects such as anisotropy, inhomogeneity, and a 90° angle of emergence.

THE ELASTICITY OF POROUS SOLIDS

In a previous paper by the author (Gassmann, 1951) the elasticity of homogeneous isotropic or anisotropic porous solids has been treated. The following investigations are based upon the results of the above mentioned paper. Therefore these results may first be summarized, all demonstrations being omitted.

A sample of a porous solid possessing the volume V and the mass m , may be composed of a homogeneous isotropic solid material of volume $\hat{V}$ and mass $\hat{m}$ and of a homogeneous liquid or gas of volume $\tilde{V}$ and mass $\tilde{m}$ filling the pores. Therefore,

$$\begin{aligned} V &= \hat{V} + \tilde{V}, & m &= \hat{m} + \tilde{m}, \\ n &= \frac{\tilde{V}}{V} = \text{porosity}. \end{aligned} \tag{1}$$

Macroscopically the porous solid may be considered as a homogeneous elastic solid if the pores are assumed sufficiently small. Though the solid material is assumed isotropic, the porous solid can behave either as an isotropic or anisotropic elastic body according to the shape of the pores. All pores are assumed to be in connection. Let us consider an initial state of stress and strain (e.g. such as the state of rocks under the weight of overlying material) and variations from this state small enough to allow the application of Hooke's law. Two extreme cases will be taken into account. In the first case the variations of stress act only upon the solid matter, the hydrostatic pressure of the pore content remaining constant, due to circulation allowed through the pores and through the surface of the body. The porous solid then behaves like an "open system" (e.g., slow variations of stress applied to a rock sample in free air). In the second case circulation of the pore content is not possible, the porous solid behaving like a "closed

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system" (e.g., stress variations in the earth's interior due to earthquake waves).
Notations:

x, y, z = rectangular coordinates

s_x, s_y, s_z = rectangular components of displacement (Love, 1944), due to stress variations.

Δe_i = components of strain variations.

$$\begin{aligned}\Delta e_1 &= \frac{\partial s_x}{\partial x}, & \Delta e_2 &= \frac{\partial s_y}{\partial y}, & \Delta e_3 &= \frac{\partial s_z}{\partial z} \\ \Delta e_4 &= \frac{\partial s_y}{\partial z} + \frac{\partial s_z}{\partial y}, & \Delta e_5 &= \frac{\partial s_z}{\partial x} + \frac{\partial s_x}{\partial z}, & \Delta e_6 &= \frac{\partial s_x}{\partial y} + \frac{\partial s_y}{\partial x}\end{aligned}\quad (2)$$

$\Delta \bar{p}_i, \Delta p_i$ = components of stress variations of the open and the closed system (pressure with positive sign).

$$\hat{\lambda}, \hat{\mu}, \hat{k} = \hat{\lambda} + \frac{2}{3} \hat{\mu} \quad (3)$$

Lame's constants and modulus of compression of the solid material.

$\tilde{k}$ = modulus of compression of the pore content. Stress-strain relations, general anisotropic case:

open system:

$$\begin{aligned}\Delta \bar{p}_i &= - \sum_{j=1}^6 \bar{c}_{ij} \Delta e_j \quad \begin{cases} i = 1, 2, \dots, 6 \\ j = 1, 2, \dots, 6 \end{cases} \\ \Delta e_i &= - \sum_{j=1}^6 \bar{\gamma}_{ij} \Delta \bar{p}_j, \quad \bar{c}_{ij} = \bar{c}_{ji}, \quad \bar{\gamma}_{ij} = \bar{\gamma}_{ji}\end{aligned}\quad (4)$$

closed system:

$$\begin{aligned}\Delta p_i &= - \sum_{j=1}^6 c_{ij} \Delta e_j, & \Delta e_i &= - \sum_{j=1}^6 \gamma_{ij} \Delta p_j \\ c_{ij} &= c_{ji}, & \gamma_{ij} &= \gamma_{ji}.\end{aligned}\quad (5)$$

Relations between the elastic constants:

$$c_{ij} = \bar{c}_{ij} + \frac{\alpha^*}{D^*} b_i b_j \quad (6)$$

$$\begin{aligned}\gamma_{ij} &= \bar{\gamma}_{ij} - \frac{\alpha^*}{D} a_i a_j \\ \alpha^* &= \frac{1}{n \left(\frac{1}{\hat{k}} - \frac{1}{\hat{k}} \right)}\end{aligned}\quad (7)$$

$$\epsilon_1 = \epsilon_2 = \epsilon_3 = 1, \quad \epsilon_4 = \epsilon_5 = \epsilon_6 = 0$$

$$a_i = \tilde{\gamma}_{i1} + \tilde{\gamma}_{i2} + \tilde{\gamma}_{i3} - \frac{\epsilon_i}{3\hat{k}}$$

$$b_i = \epsilon_i - \frac{1}{3\hat{k}}(\tilde{c}_{i1} + \tilde{c}_{i2} + \tilde{c}_{i3})$$

$$D = 1 + \alpha^*(a_1 + a_2 + a_3)$$

$$D^* = 1 + \frac{\alpha^*}{3\hat{k}}(b_1 + b_2 + b_3).$$

In particular, for the isotropic case:

$$\bar{\lambda}, \bar{\mu}, \bar{k} = \bar{\lambda} + \frac{2}{3} \bar{\mu} \quad \text{elastic constants of the open system.}$$

$$\lambda, \mu, k = \lambda + \frac{2}{3} \mu \quad \text{elastic constants of the closed system.}$$

From (7) is obtained

$$\begin{aligned} \mu &= \bar{\mu} & k &= \hat{k} \cdot \frac{\bar{k} + Q}{\hat{k} + Q} \\ Q &= \frac{\hat{k}(\hat{k} - \bar{k})}{n(\hat{k} - \tilde{k})} \end{aligned} \quad (8)$$

HEXAGONAL CLOSE PACKING OF SPHERES, HOMOGENEOUSLY STRESSED

Let two homogeneous and isotropic spheres of equal radius R (with Lamé's constants $\hat{\lambda}, \hat{\mu}$) be in contact under the pressure P (Figure 1).

$$\hat{E} = \frac{\hat{\mu}(3\hat{\lambda} + 2\hat{\mu})}{\hat{\lambda} + \hat{\mu}} = \text{Young's modulus,}$$

$$\hat{\nu} = \frac{\hat{\lambda}}{2(\hat{\lambda} + \hat{\mu})} = \text{Poisson's ratio.}$$

According to Hertz's theory (Love, 1944), the radius of the circle of contact is

$$R' = \sqrt[3]{\frac{3(1 - \hat{\nu})^2 R}{4\hat{E}}} P. \quad (9)$$

R' is assumed to be small compared with R , and

$$s \cong \frac{R'^2}{R} = \sqrt[3]{\frac{9(1 - \hat{\nu})^2 P^2}{16\hat{E}^2 R}} = \text{relative displacement of the spheres.} \quad (10)$$

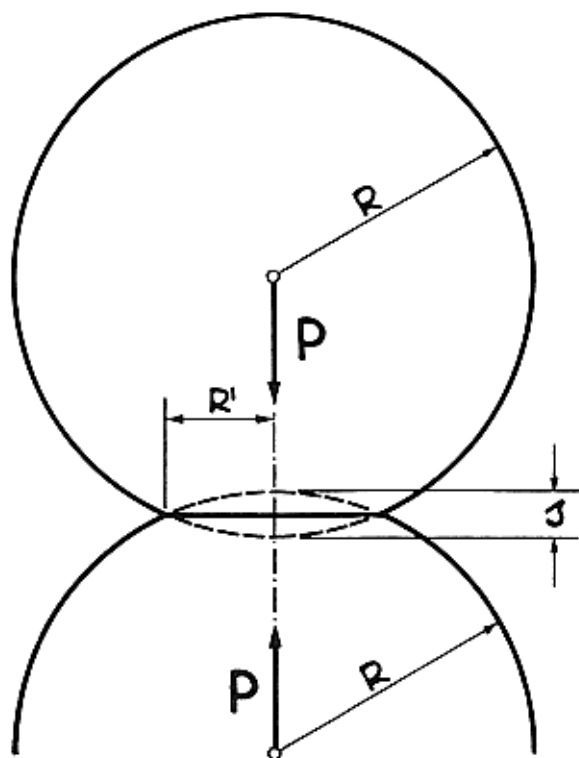


FIG. 1. Two equal spheres in contact.

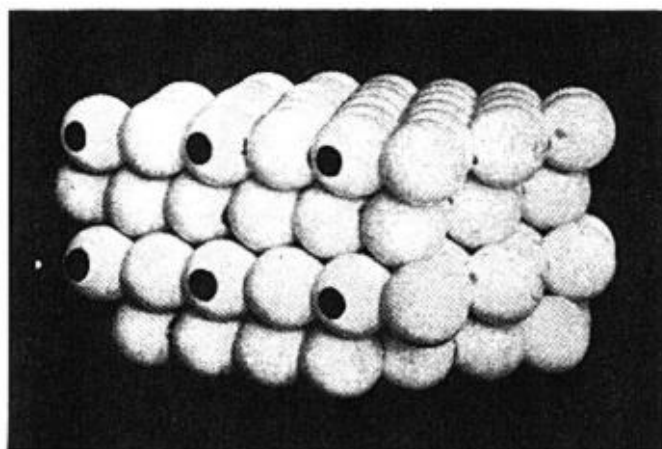


FIG. 2. The hexagonal close packing of spheres.

The hexagonal close packing of equal spheres (Figure 2) now to be considered (following an idea of Hara, 1935) consists of different layers (Figure 3). The coordinates of the centers of the spheres belonging to the N th layer (no pressure assumed) are

$$\begin{aligned} x &= \frac{R}{2} [4N' + 2N'' + (-1)^N + 1] \\ y &= \frac{\sqrt{3}R}{4} [4N'' + (-1)^N + 1] \\ z &= \frac{2\sqrt{6}}{3} RN, \quad N' \text{ and } N'' \text{ being arbitrary real integers.} \end{aligned} \quad (11)$$

As initial stress, we assume zero pressure between the spheres of the same layer,

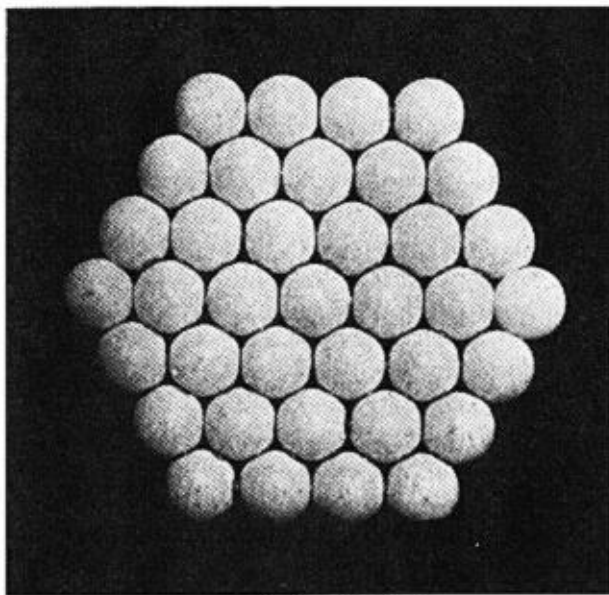


Fig. 3. A layer of spheres of the hexagonal close packing.

i.e., the spheres with centres of the same value of z , and a pressure $P > 0$ between each pair of spheres in contact, belonging to neighboring layers. The corresponding slight deformation of the lattice (11) of the centers can be neglected in the following considerations. If the number of layers and spheres is sufficiently large, the packing at its initial state of stress can be considered as a homogeneous anisotropic porous solid

$$\left[\text{porosity } n = 1 - \frac{\pi}{3\sqrt{2}} = 0.260 \right] \quad (12)$$

with the symmetry of Voigt's class Nr. 26 of crystals. With respect to elasticity all directions perpendicular to z are equivalent ("transversely isotropic" symmetry, Love 1944, page 160) and the elasticity is determined by five constants $C_1, C_2, \dots, C_5$, the matrix of the open system constants $\bar{c}_{ij}$ in (4) taking the form

$$\left\{ \begin{array}{cccccc} C_1 & C_1 - 2C_5 & C_2 & 0 & 0 & 0 \\ C_1 - 2C_5 & C_1 & C_2 & 0 & 0 & 0 \\ C_2 & C_2 & C_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & C_4 & 0 & 0 \\ 0 & 0 & 0 & 0 & C_4 & 0 \\ 0 & 0 & 0 & 0 & 0 & C_5 \end{array} \right\}. \quad (13)$$

The $\bar{c}_{ij}$ can be taken from (4), if for suitable homogeneous deformations of the lattice (11) the variations of stress are calculated by means of (10), variations Δs and ΔP being replaced by ds and dP . With

$$\bar{c} = \frac{1}{12\sqrt{2}} \cdot \frac{1}{R} \cdot \frac{dP}{ds} = \frac{1}{12\sqrt{2}} \cdot \sqrt[3]{\frac{6E^2P}{(1 - \hat{\nu}^2)^2R^2}} \quad (14)$$

the matrix $\bar{c}_{ij}$ is obtained by putting

$$C_1 = \bar{c}, \quad C_2 = C_4 = 4\bar{c}, \quad C_3 = 16\bar{c}, \quad C_5 = 0 \quad (15)$$

into (13).

THE PACKING OF SPHERES STRESSED BY SELF-WEIGHT

The packing of spheres introduced by Figures 2 and 3 and the formulas (11) may now be considered under the influence of self-weight, the $+z$ axis pointing vertically downward. The interspaces between the spheres may be filled with a liquid of constant density $\tilde{\rho}$. Each sphere belonging to the N th layer is in contact with three spheres of the $(N+1)$ th layer under the pressure P_N and with three spheres of the $(N-1)$ th layer under the pressure P_{N-1} . The condition for equilibrium of such a sphere contains the vertical components of the six pressures, and the apparent weight $4\pi/3 R^3(\hat{\rho} - \tilde{\rho})g$ of the sphere when $g = 980.665 \text{ cm sec}^{-2}$. It is

$$P_{N-1} \cdot \sqrt{6} - P_N \cdot \sqrt{6} + \frac{4\pi}{3} R^3(\hat{\rho} - \tilde{\rho})g = 0. \quad (16)$$

If no load is assumed above the first layer, then $P_0 = 0$ and the recurrence formula (16) leads to

$$P_N = N \cdot \frac{4\pi}{3\sqrt{6}} R^3(\hat{\rho} - \tilde{\rho})g \quad (17)$$

or taking the value of N from (11):

$$P_N = \frac{\pi}{3} R^2 (\hat{\rho} - \tilde{\rho}) g z. \quad (18)$$

Substituting this expression for P in (14), we obtain

$$\bar{c} = \frac{1}{12\sqrt{2}} \sqrt[3]{\frac{2\pi\hat{E}^2(\hat{\rho} - \tilde{\rho})g}{(1 - \hat{\nu}^2)^2}} \sqrt[3]{z}. \quad (19)$$

If the pores are filled with a liquid, the packing of spheres may be called a "wet system." If the pores are evacuated [$\tilde{\rho} = 0$], the packing may be called a "dry system." If the pores are filled with air, its density and its modulus of compression are negligible. Therefore the system possesses the elasticity of a dry system.

$$\bar{\rho} = (1 - n)\hat{\rho} = \text{density of the dry system.} \quad (20)$$

$$\rho = (1 - n)\hat{\rho} + n\tilde{\rho} = \text{density of the wet system.}$$

$$(n = 0.26, \text{ see (12)}).$$

The stress below the plane $z = z_0 > 0$ (z_0 large compared with R) is not changed if the packing of spheres between the planes $z = 0$ and $z = z_0$ is replaced by an arbitrary layer between the planes $z = z_1$ and $z = z_0$, with density ρ^* , depending only on z , if

$$\int_{z_1}^{z_0} \rho^*(z) dz = \rho z_0. \quad (21)$$

The packing of spheres may be considered as a simple model for a gravel or sand, dry (= dry system) or filled with ground water (= wet system).

PROPAGATION OF ELASTIC WAVES THROUGH THE PACKING OF SPHERES

The elasticity of the dry system is given by the matrix of the $\bar{c}_{ij}$ which is calculated by means of (13), (15) and (19) with $\tilde{\rho} = 0$. In order to calculate the elastic constants of the closed wet system, first the matrix of the $\bar{c}_{ij}$ is calculated in the same way, but with $\tilde{\rho} \neq 0$ in $\bar{c}$ of (19). Then the matrix of the c_{ij} , found by (6), is the matrix (13) with

$$\begin{aligned} C_1 &= \bar{c} + \frac{\alpha^*}{D^*} b_1^2, & C_2 &= 4\bar{c} + \frac{\alpha^*}{D^*} b_1 b_3, \\ C_3 &= 16\bar{c} + \frac{\alpha^*}{D^*} b_3^2, & C_4 &= 4\bar{c}, & C_5 &= 0, \end{aligned} \quad (22)$$

$$\left(b_1 = b_2 = 1 - 2 \frac{\bar{c}}{k}, \quad b_3 = 1 - 8 \frac{\bar{c}}{k}, \quad b_4 = b_5 = b_6 = 0 \right).$$

The partial differential equations of wave motion, the wave length assumed

large compared with R , are obtained by substituting the stress-strain relations into the general equations of motion of an elastic solid (Love, 1944):

$$\begin{aligned}
 \rho \frac{\partial^2 s_x}{\partial t^2} &= C_1 \frac{\partial^2 s_x}{\partial x^2} + C_5 \frac{\partial^2 s_x}{\partial y^2} + C_4 \frac{\partial^2 s_x}{\partial z^2} + (C_1 - C_5) \frac{\partial^2 s_y}{\partial x \partial y} \\
 &\quad + (C_2 + C_4) \frac{\partial^2 s_z}{\partial x \partial z}, \\
 \rho \frac{\partial^2 s_y}{\partial t^2} &= C_5 \frac{\partial^2 s_y}{\partial x^2} + C_1 \frac{\partial^2 s_y}{\partial y^2} + C_4 \frac{\partial^2 s_y}{\partial z^2} + (C_2 + C_4) \frac{\partial^2 s_z}{\partial y \partial z} \\
 &\quad + (C_1 - C_5) \frac{\partial^2 s_x}{\partial x \partial y}, \\
 \rho \frac{\partial^2 s_z}{\partial t^2} &= C_4 \frac{\partial^2 s_z}{\partial x^2} + C_4 \frac{\partial^2 s_z}{\partial y^2} + C_3 \frac{\partial^2 s_z}{\partial z^2} + (C_2 + C_4) \frac{\partial^2 s_x}{\partial x \partial z} \\
 &\quad + (C_2 + C_4) \frac{\partial^2 s_y}{\partial y \partial z},
 \end{aligned} \tag{23}$$

where t =time. For the wet system, the C_i 's are to be taken from (22), for the dry system from (15), putting $\tilde{p}=0$ in (19) and (20). Now let $W(x, y, z, t)$ = const. = equation of a wave front.

$$\frac{\partial W}{\partial t} = q_0, \quad \frac{\partial W}{\partial x} = q_1, \quad \frac{\partial W}{\partial y} = q_2, \quad \frac{\partial W}{\partial z} = q_3. \tag{24}$$

The characteristic relation (Levi-Civita, 1932) of the system (23) is

$$\begin{vmatrix} A_{11} - \rho q_0^2 & A_{12} & A_{13} \\ A_{12} & A_{22} - \rho q_0^2 & A_{23} \\ A_{13} & A_{23} & A_{33} - \rho q_0^2 \end{vmatrix} = 0, \tag{25}$$

$$\begin{aligned}
 A_{11} &= C_1 q_1^2 + C_5 q_2^2 + C_4 q_3^2, & A_{12} &= (C_1 - C_5) q_1 q_2, \\
 A_{22} &= C_5 q_1^2 + C_1 q_2^2 + C_4 q_3^2, & A_{23} &= (C_2 + C_4) q_2 q_3, \\
 A_{33} &= C_4 q_1^2 + C_4 q_2^2 + C_3 q_3^2, & A_{13} &= (C_2 + C_4) q_1 q_3.
 \end{aligned}$$

Assuming a point source (=focus) at $x=y=z=0$ and considering the wave propagation in the (x,z) plane, we have by reason of symmetry, $q_2=0$

$$v = \left| \frac{q_0}{\sqrt{q_1^2 + q_3^2}} \right| = \text{velocity of propagation perpendicular to the wave front.} \tag{26}$$

$\tan \alpha = \frac{q_1}{q_3}$, α = angle between the normal of the wave front and the z axis.

The three roots of (25) corresponding to the three different kinds of waves with velocities $v=v_1, v_2, v_3$ lead to

$$\left. \begin{aligned} v_1^2 \right\} &= \frac{1}{2\rho(q_1^2 + q_3^2)} (A_{11} + A_{33} \pm \sqrt{A_{11}^2 + A_{33}^2 + 4A_{13}^2 - 2A_{11}A_{33}}) \\ v_2^2 &= \frac{A_{22}}{\rho(q_1^2 + q_3^2)}. \end{aligned} \quad (27)$$

As an example, the velocity $v_1 (v_1 > v_2 > v_3)$ has been calculated for a packing of granitic spheres, dry [$v_1 = v_1(z, \alpha)$] and filled with water [$v_1 = v_1^*(z, \alpha)$] for $\alpha = 0^\circ$ (vertical direction of propagation) and $\alpha = 90^\circ$ (horizontal direction), using the following values:

$$\begin{aligned} \hat{\rho} &= 2.65 \frac{\text{gr}}{\text{cm}^3}, & \hat{E} &= 0.5 \cdot 10^{12} \frac{\text{dyne}}{\text{cm}^2}, & \hat{\nu} &= 0.25, \\ \tilde{\rho} &= \begin{cases} 0 & \text{dry,} \\ 1 & \text{gr wet,} \end{cases} & \tilde{k} &= 2.06 \cdot 10^{10} \frac{\text{dyne}}{\text{cm}^2}. \end{aligned} \quad (28)$$

The results are shown in Figure 4.

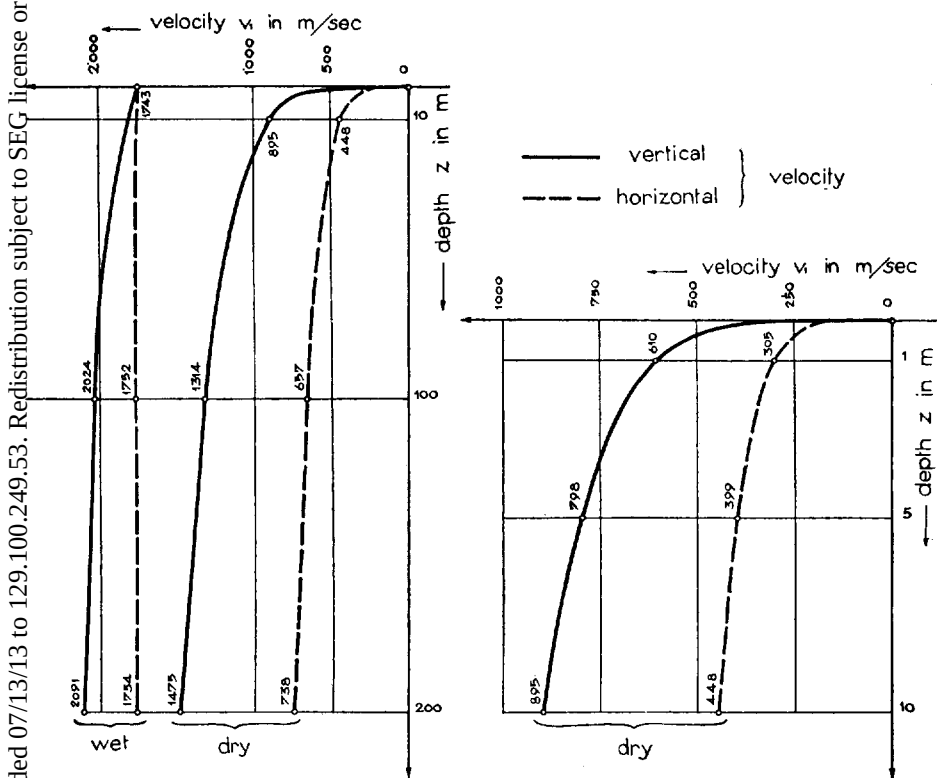


FIG. 4. Velocities of wave propagation in a packing of spheres.

The wave rays (= seismic rays) are the bi-characteristics of (23). For the rays in the (x, z) plane ($q_2 = 0$), the differential equations are:

$$\begin{aligned} \frac{dx}{dt} &= \frac{\partial H}{\partial q_1}, & \frac{dz}{dt} &= \frac{\partial H}{\partial q_3}, \\ \frac{dq_1}{dt} &= -\frac{\partial H}{\partial x}, & \frac{dq_3}{dt} &= -\frac{\partial H}{\partial z}, & \frac{dq_0}{dt} &= -\frac{\partial H}{\partial t}. \end{aligned} \quad (29)$$

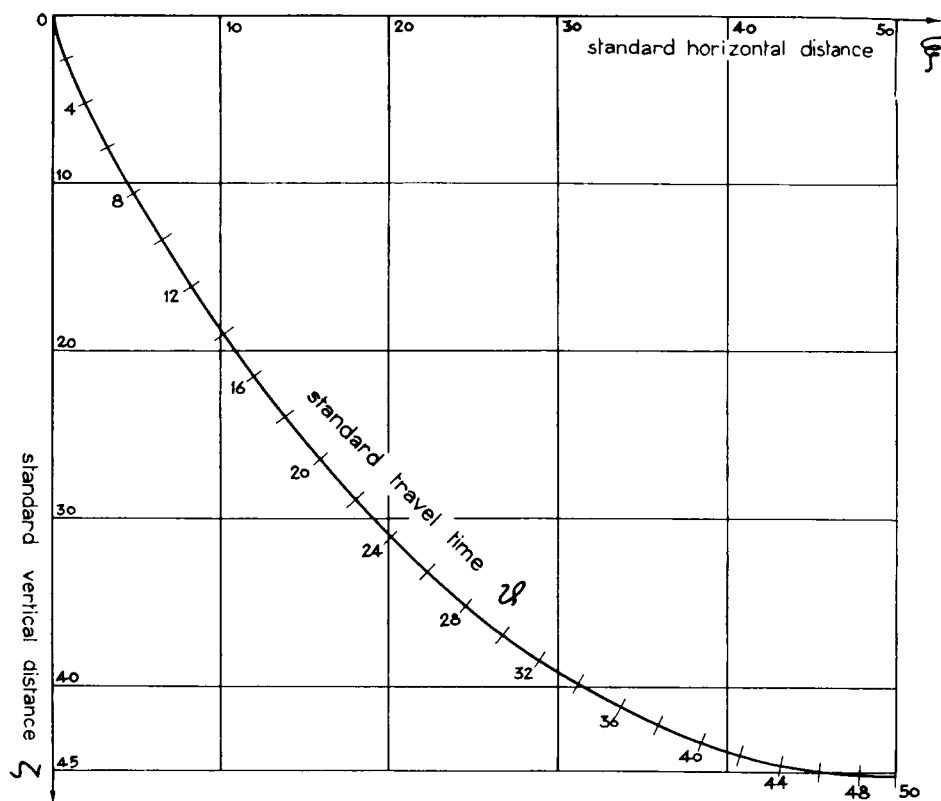


FIG. 5. Standard ray of elastic waves in a packing of spheres.

H , corresponding e.g., to v_1 , is obtained by taking the square root of the corresponding equation (27):

$$q_0 = -H(z, q_1, q_3). \quad (30)$$

The rays corresponding to v_1 will now be calculated for the dry system:

$$H = \frac{L}{4\sqrt{2}} \cdot \sqrt[6]{z} \sqrt{5(q_1^2 + 4q_3^2) + \sqrt{9q_1^4 + 328q_1^2q_3^2 + 144q_3^4}},$$

$$L = \sqrt[3]{\frac{8E\sqrt{2g}}{\pi\hat{p}(1 - \hat{v}^2)}}.$$
(31)

We introduce α as parameter

$$F(\alpha) = \frac{1}{8} \sqrt{25 + 15 \cos 2\alpha + \sqrt{(15 + 9 \cos 2\alpha)^2 + 256 \sin^2 2\alpha}},$$

$$\frac{dF}{d\alpha} = F'(\alpha),$$

$$v_1 = L \cdot \sqrt[6]{z} \cdot F(\alpha),$$

$$H = L \cdot \sqrt[6]{z} \cdot \frac{F(\alpha)}{\sin \alpha} \cdot q_1$$
(32)

and integrate (29).

If $t=0$ is assumed as focal time, the results are:

$$T(\alpha) = 5 \int_0^\alpha \frac{\sin^4 \psi}{[F(\psi)]^6} d\psi$$
(33)

(auxiliary function, graphically computed)

$$T_1 = T(\pi) = 141,950$$
(34)

(integration by aid of elliptic integrals)

standard ray (Figure 5):

$$\xi = \frac{100}{T_1} \left\{ T(\alpha) - \left[\frac{\sin \alpha}{F(\alpha)} \right]^6 \cot \alpha \right\}$$

$$\zeta = \frac{100}{T_1} \left[\frac{\sin \alpha}{F(\alpha)} \right]^6$$

$$\vartheta = \frac{100}{T_1} \cdot T(\alpha) \quad (= \text{standard travel time})$$
(35)

angle of emergence = 90°

Arbitrary ray:

a = horizontal focal distance of the ray.

$$x = \frac{a}{100} \xi, \quad z = \frac{a}{100} \zeta,$$

$$t = 0.012 \sqrt{T_1} \cdot \frac{a^{5/6}}{L} \cdot \vartheta = 0.027408 \cdot \frac{a^{5/6}}{L} \cdot \vartheta.$$
(36)

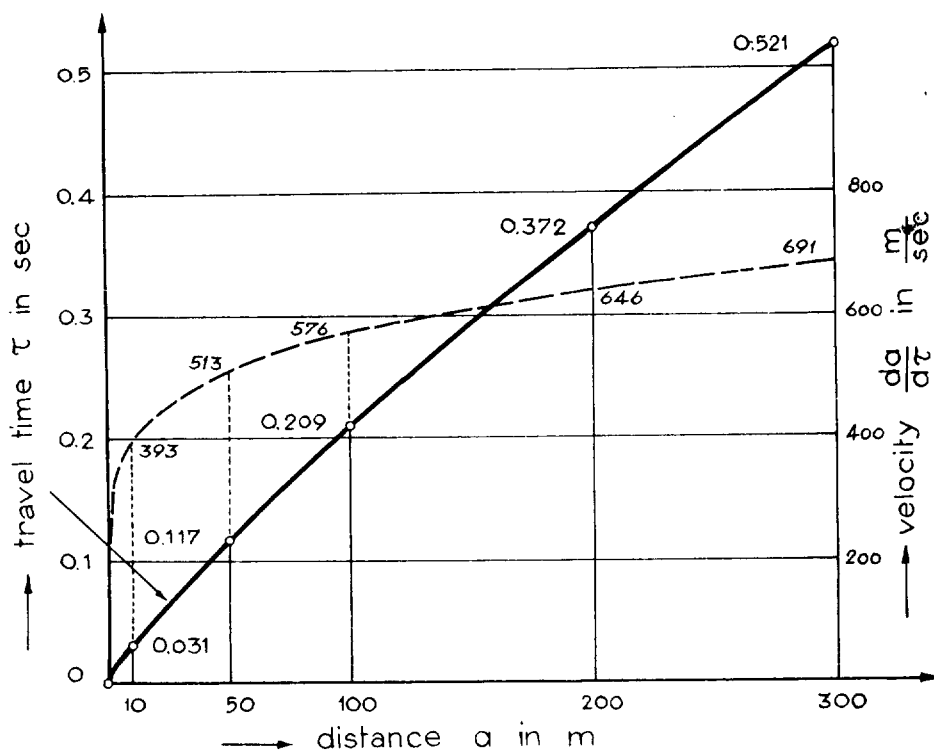


FIG. 6. Travel time and apparent velocity of elastic waves in a packing of spheres.

The ray is symmetric with respect to $x = a/2$.

Travel time function:

$$\alpha = \pi, \quad x = a, \quad z = 0, \quad t = \tau = \frac{2.7408}{L} a^{5/6} \quad (\text{Figure 6}). \quad (37)$$

Deepest point of the ray:

$$\alpha = \frac{\pi}{2}, \quad x = \frac{a}{2}, \quad z = z_s = \frac{64}{T_1} a = 0.451a. \quad (38)$$

ACKNOWLEDGMENTS

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